

Chap. 9

Object Interaction

Object Oriented Systems Analysis and Design Using UML, (4th Edition),
McGraw Hill

Topics Covered

- Sequence Diagrams
- Communications Diagrams
- Model Consistency

Modelling Interactions

- Interactions can be modelled using various notations
 - Sequence diagrams
 - Communication diagrams
 - Interaction overview diagrams
 - Timing diagrams

Sequence Diagrams

- Shows an interaction between objects arranged in a time sequence.
- Typically used to represent the detailed object interaction that occurs for one use case.

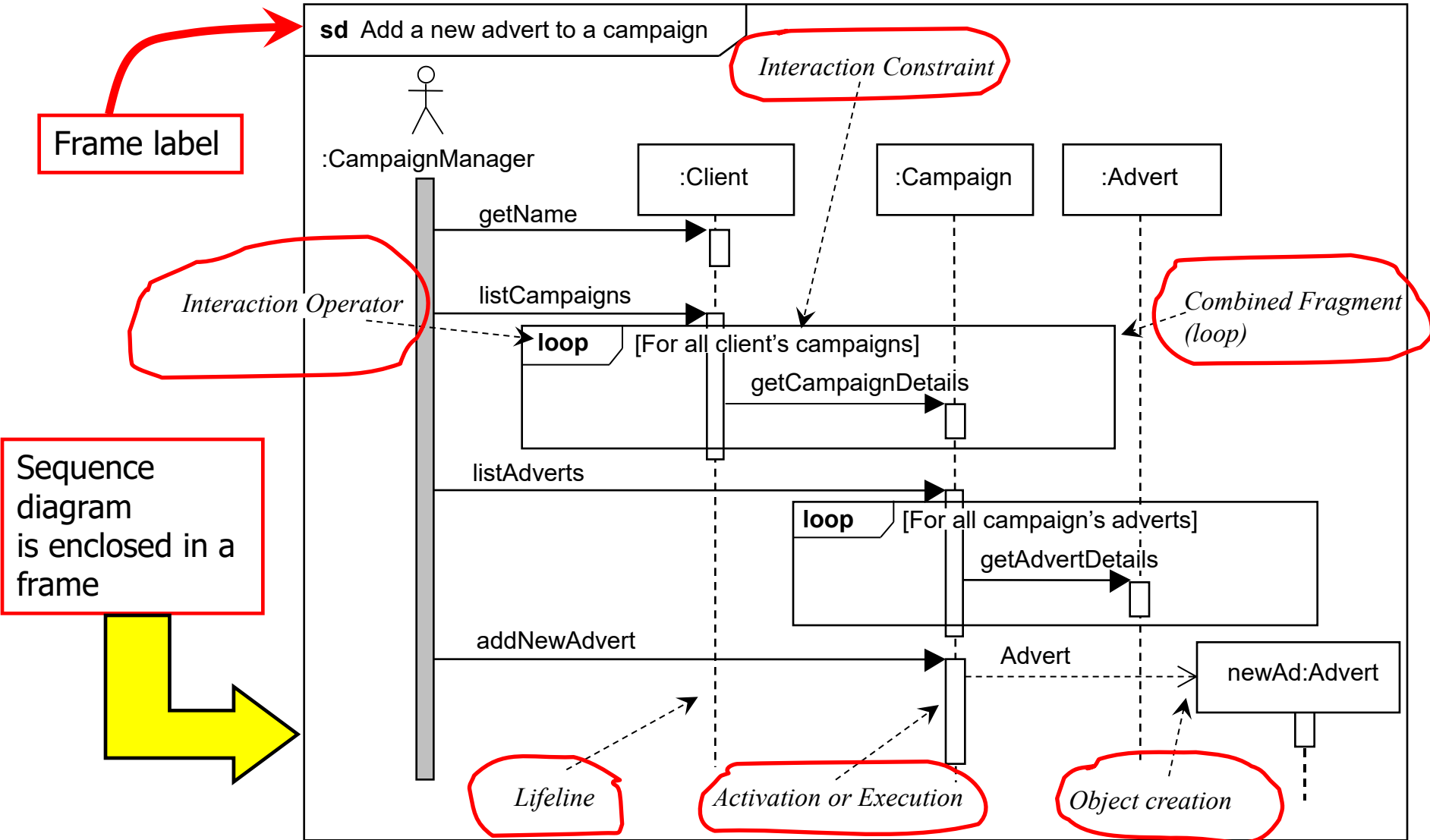
Sequence Diagrams

- Vertical dimension
 - shows time
- Objects (or subsystems or other connectable objects) involved in interaction
 - appear horizontally across the page and are represented by lifelines
- Messages
 - shown by a solid horizontal arrow
- The execution or activation of an operation
 - shown by a rectangle on the relevant lifeline

Sequence Diagram

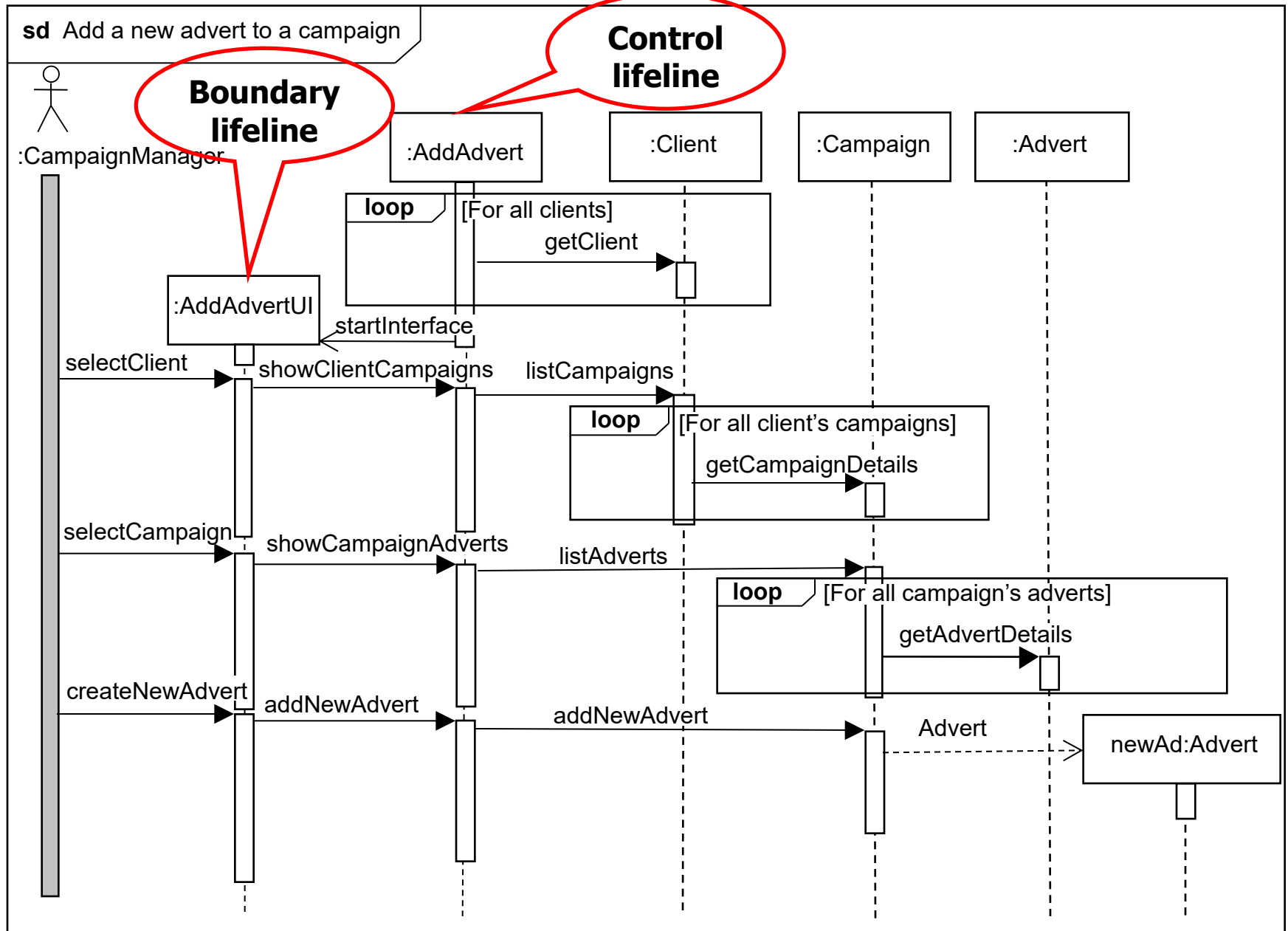
- Iteration
 - represented by *combined fragment* rectangle with the *interaction operator* 'loop'.
- The loop combined fragment
 - executed only if the guard condition in the interaction constraint evaluates as true.
- Object creation
 - shown with the construction arrow (dashed) going to the object symbol for the Advert lifeline.

Sequence diagram



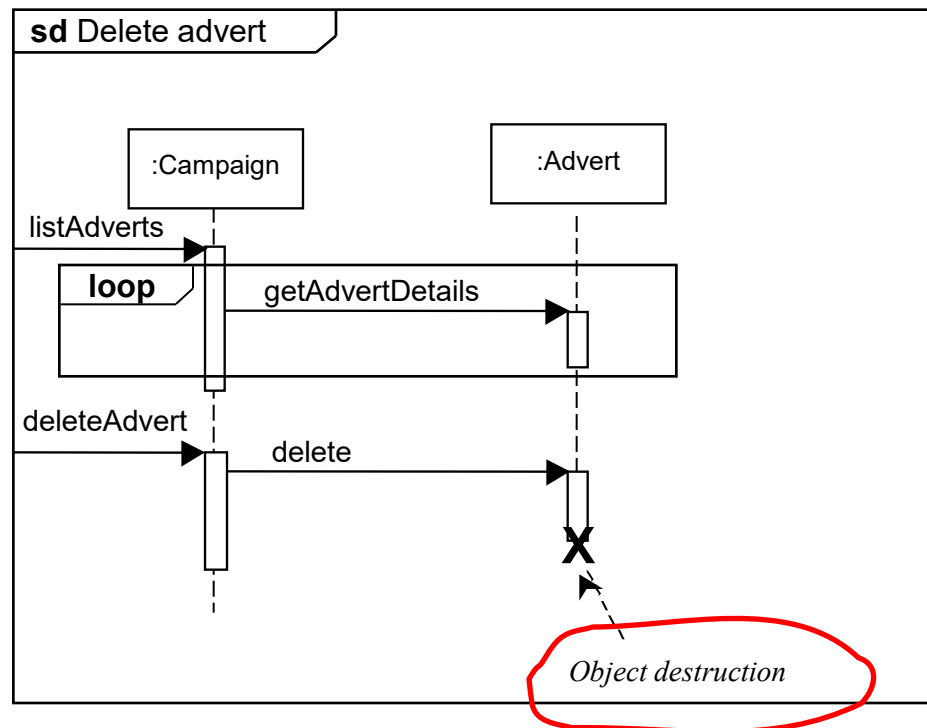
Boundary & Control Classes

- A use case includes at least
 - one boundary object
 - manages the dialogue between the actor and the system
 - represented by the lifeline `:AddAdvertUI`
 - one control object
 - manages the overall object communication
 - represented by the lifeline `:AddAdverts`



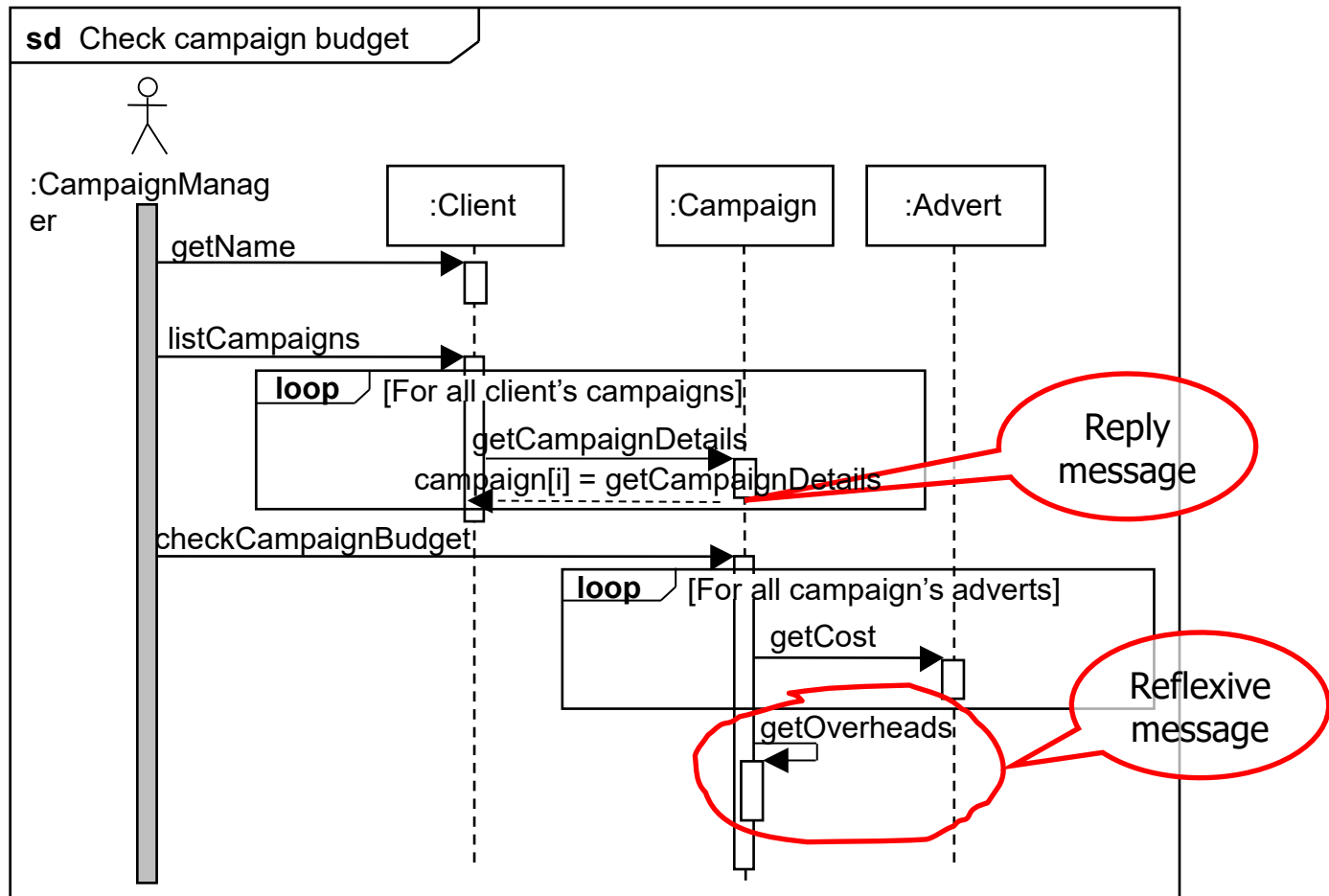
Object Destruction

- Indicated by a large X on the lifeline at the point in the interaction



Reply & Reflexive Messages

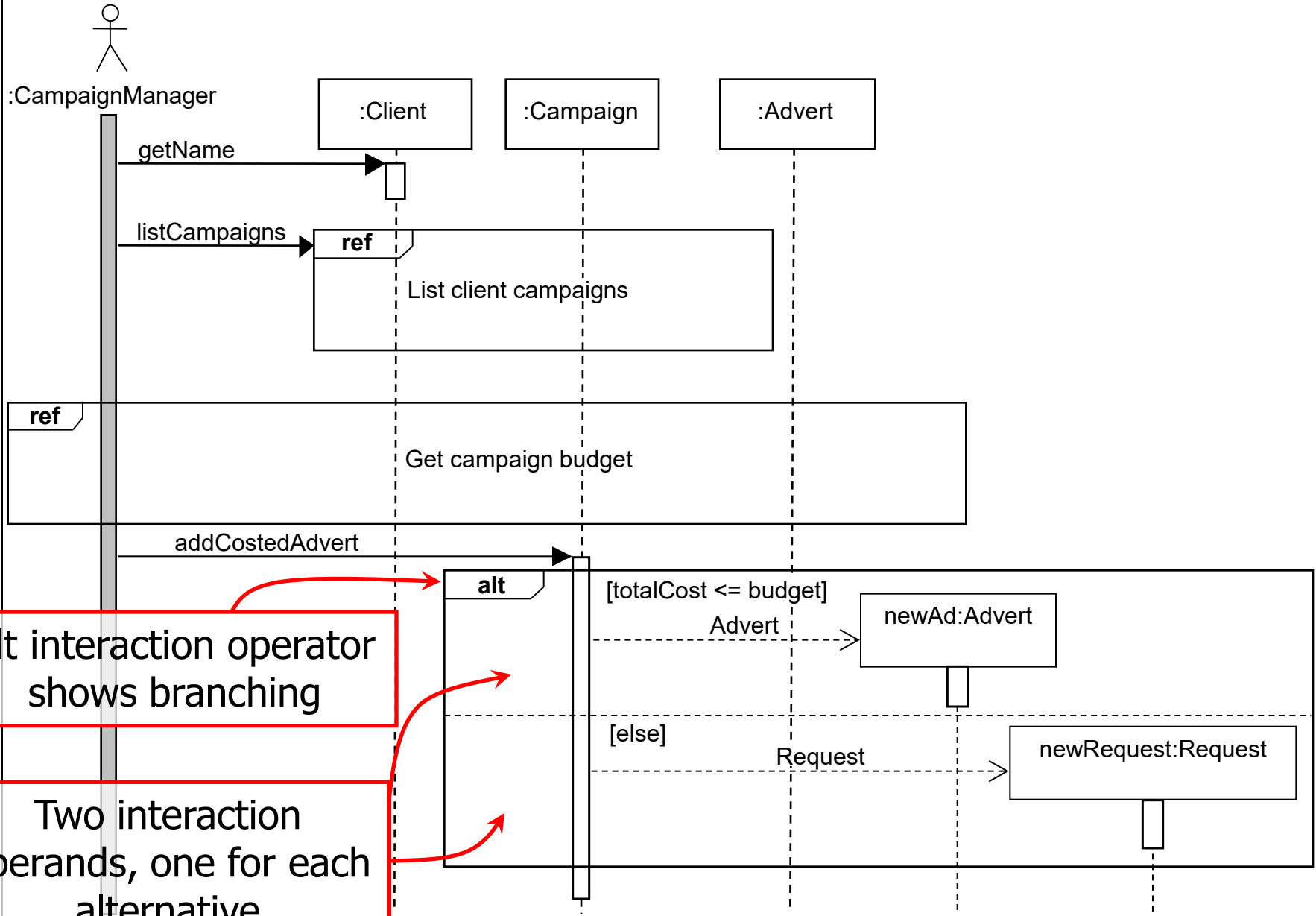
- A reply message returns the control to the object that originated the message, but it is optional
- A reflexive message is the one that an object send to itself



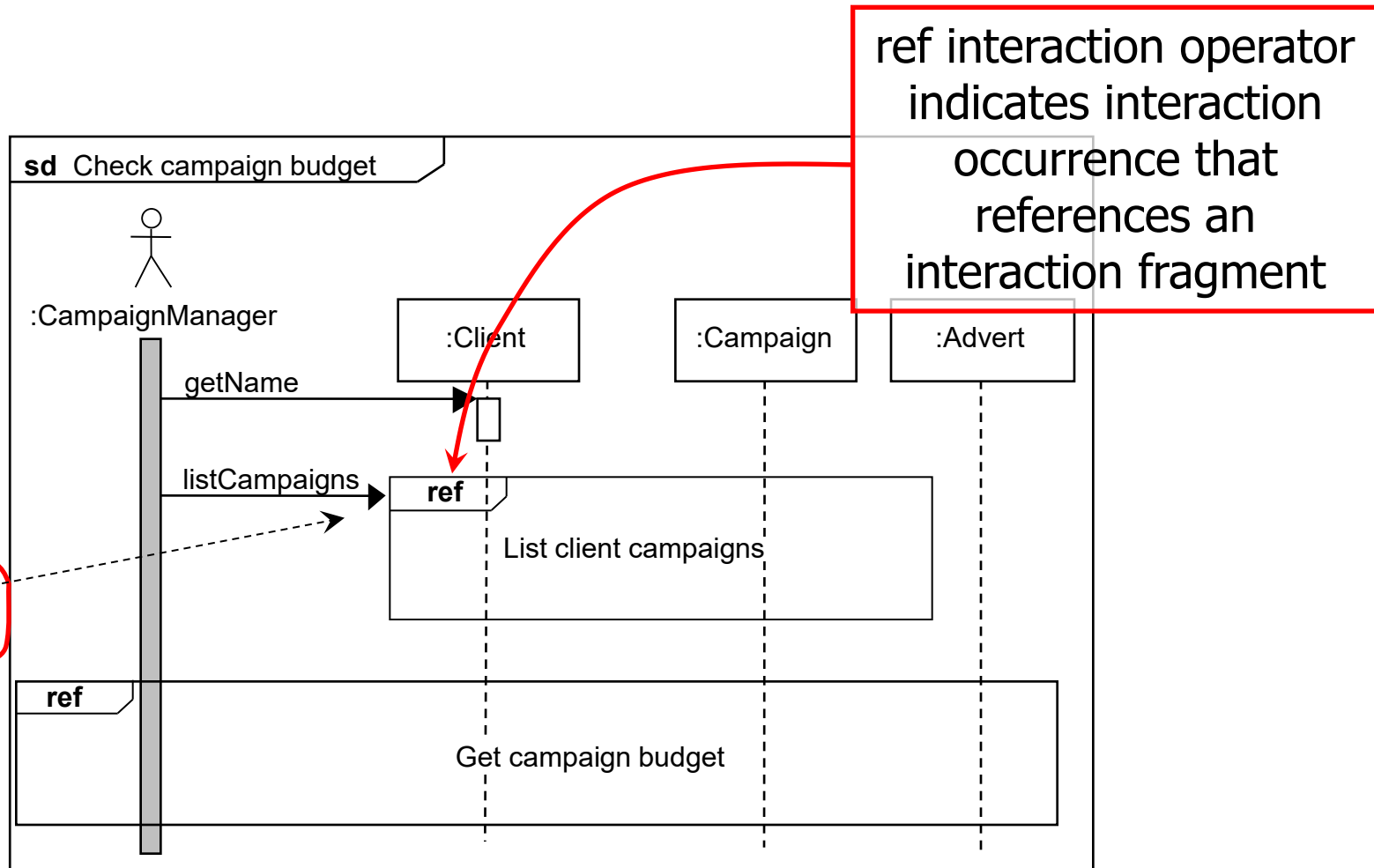
Handling Complexity

- Complex interactions can be modelled using various different techniques
 - Interaction fragments
 - Lifelines for subsystems or groups of objects

sd Add a new advert to a campaign if within budget

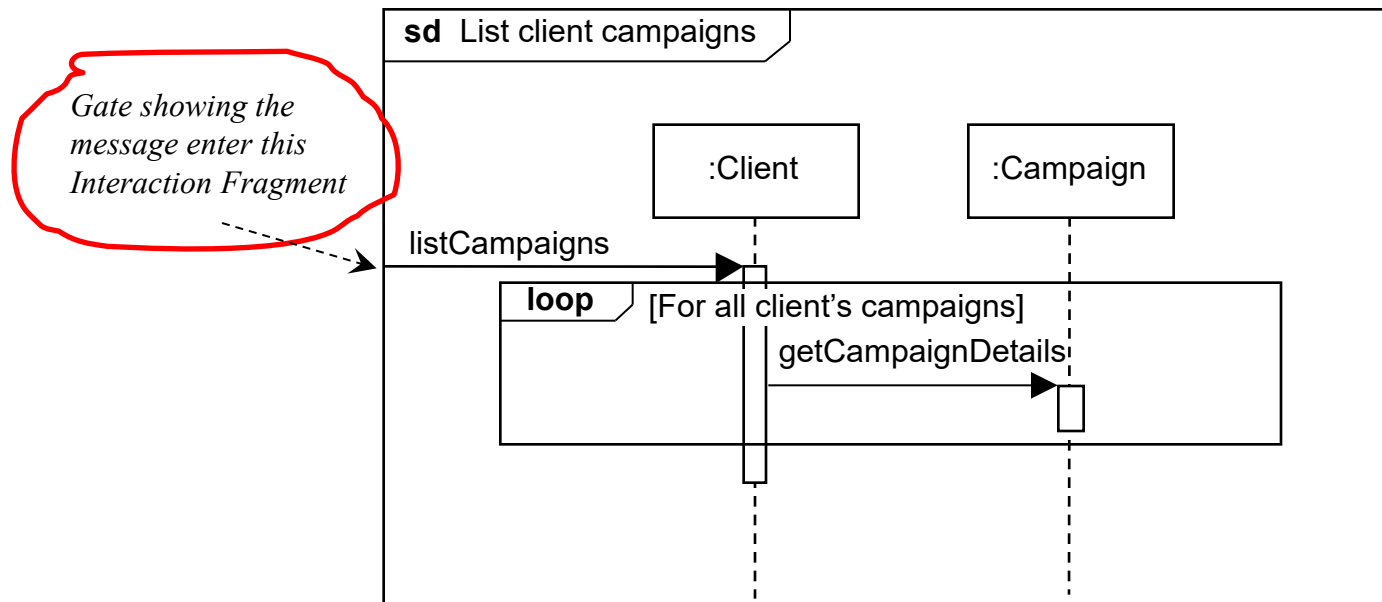


Using Interaction Fragments



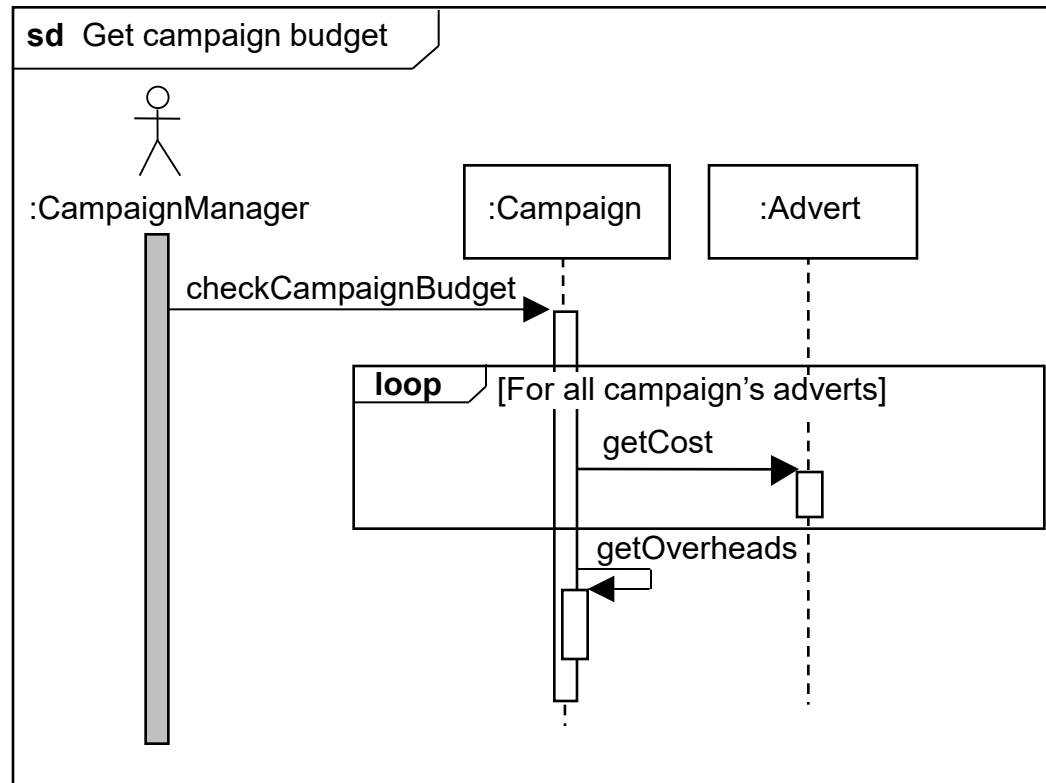
Interaction Fragment

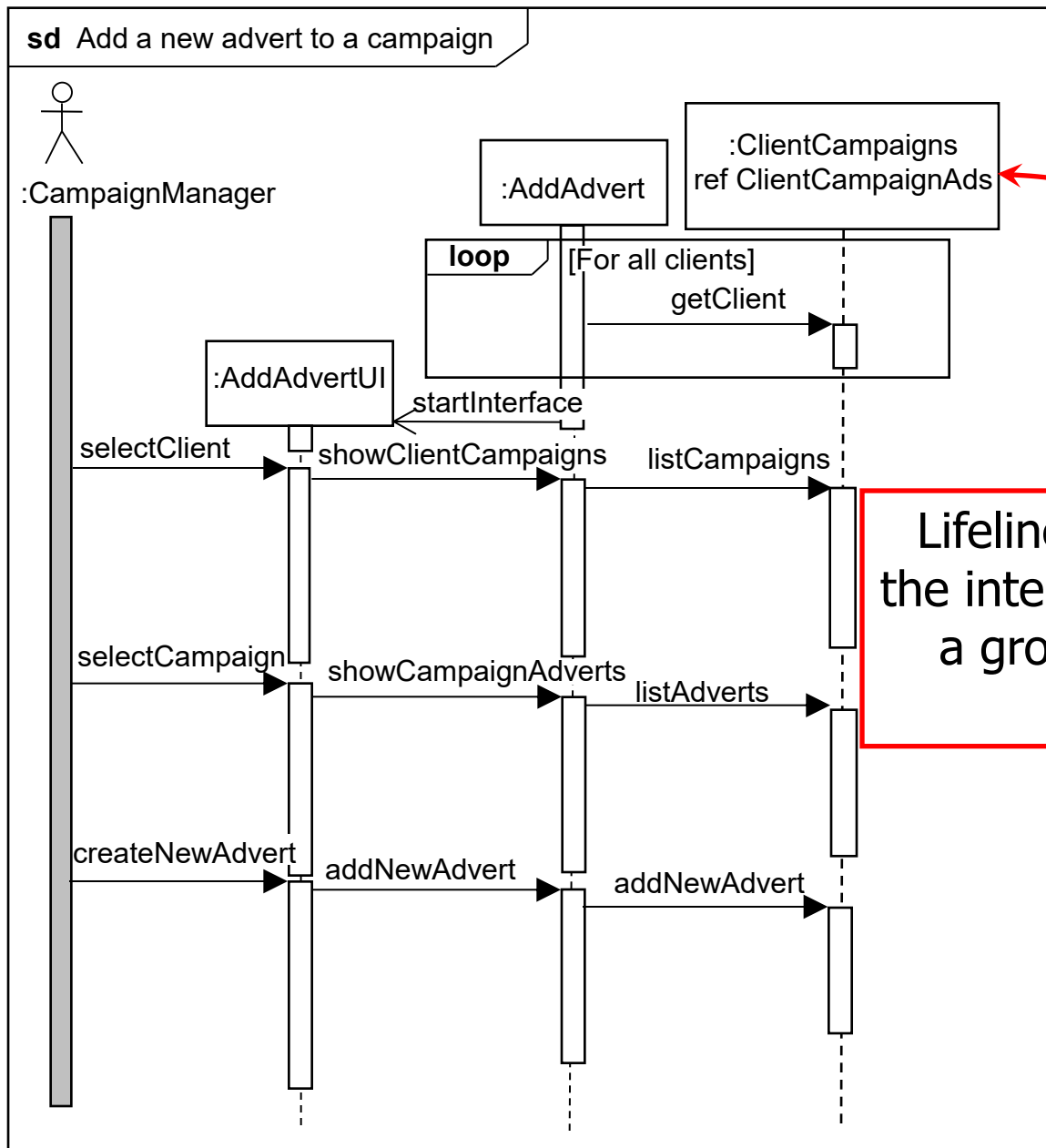
- Reusable by other actors



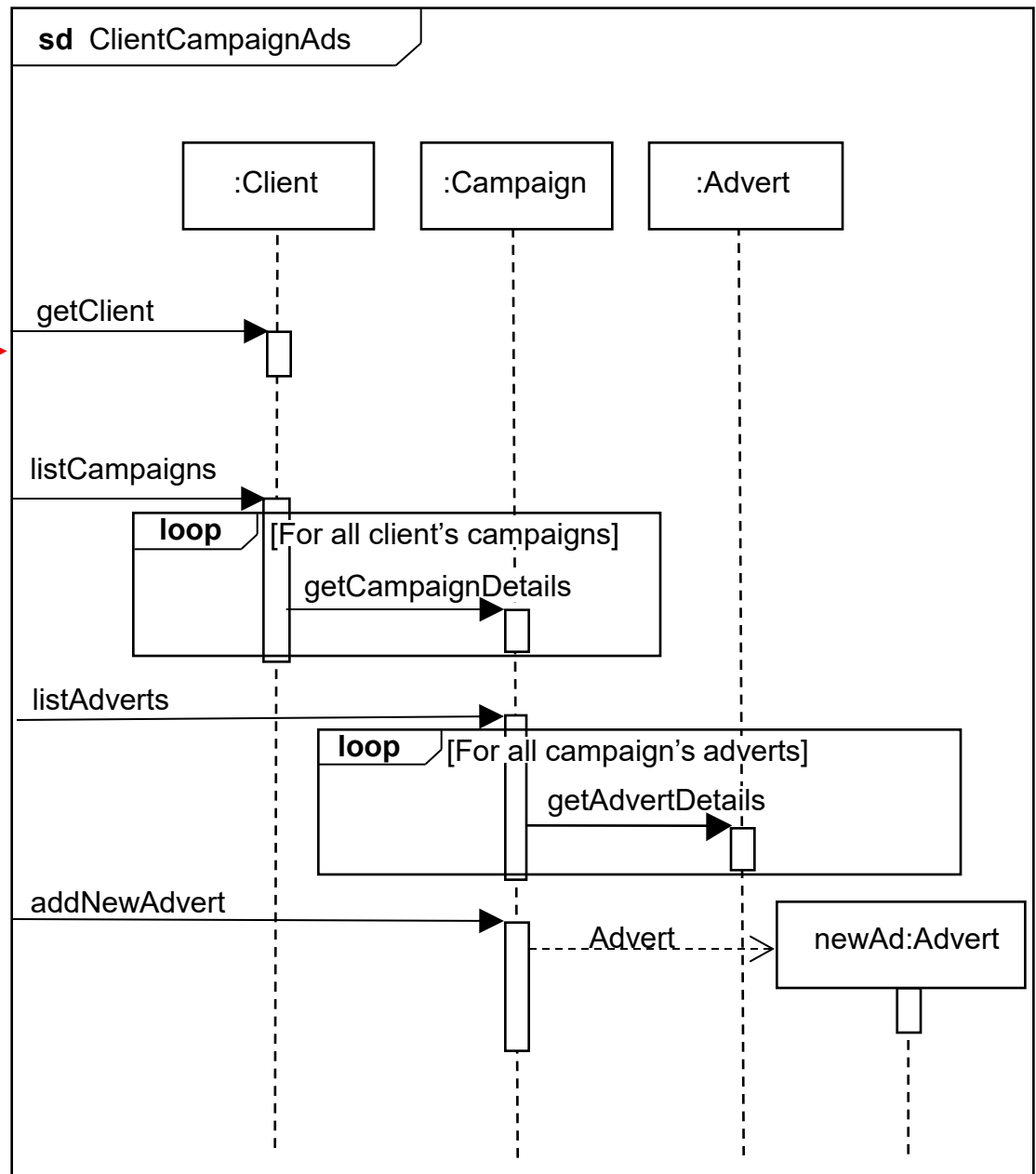
Interaction Fragment

- Only actor 'Campaign Manager can use this interaction





Sequence diagram
referenced in the
Add a new advert
to a campaign
sequence diagram



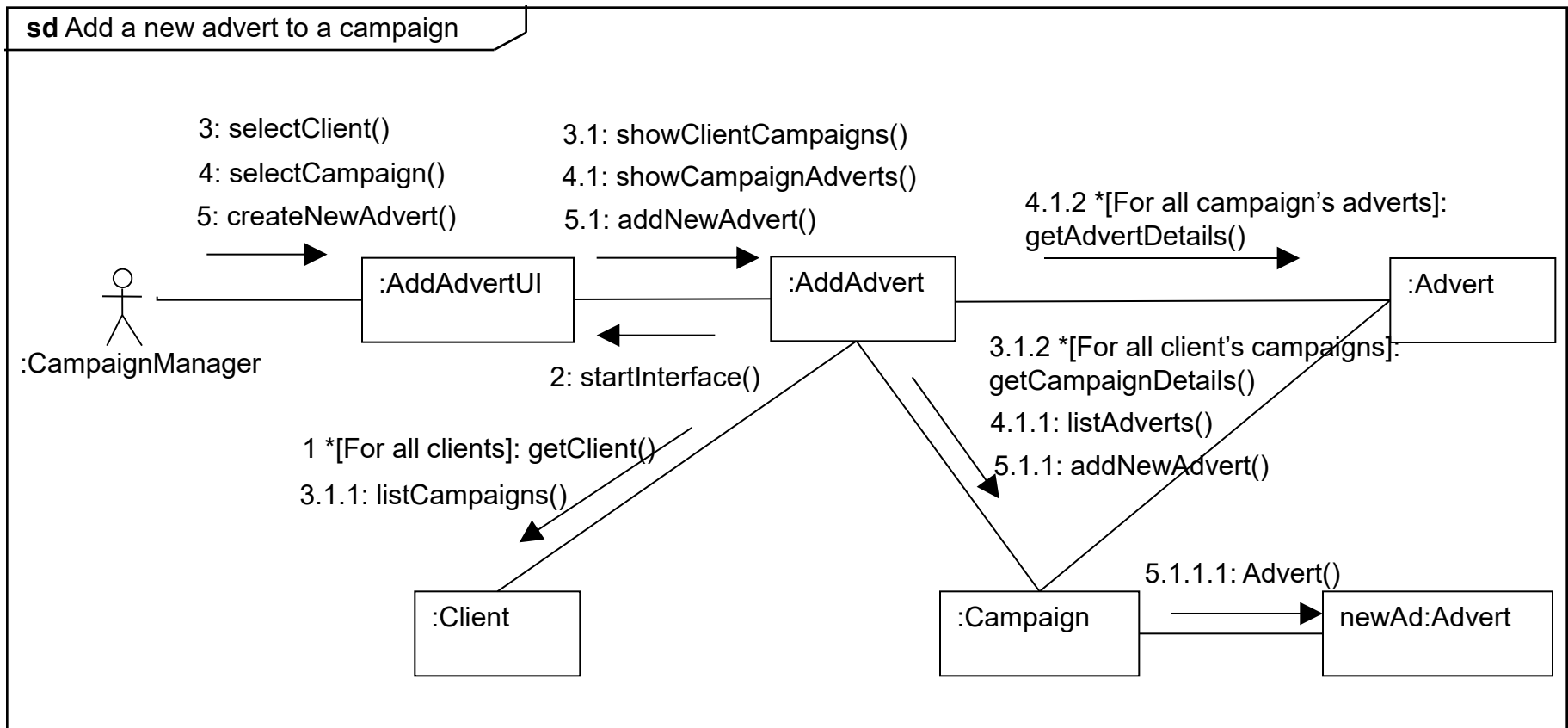
Interaction Operators

Interaction Operator	Explanation and use
<code>alt</code>	Alternatives represents alternative behaviours, each choice of behaviour being shown in a separate operand. The operand whose interaction constraint is evaluated as true executes.
<code>opt</code>	Option describes a single choice of operand that will only execute if its interaction constraint evaluates as true.
<code>loop</code>	Loop is used to indicate an operand that is repeated a number of times until the interaction constraint for the loop is no longer true.
<code>break</code>	Break indicates that the combined fragment is performed instead of the remainder of the enclosing interaction fragment.
<code>par</code>	Parallel indicates that the execution operands in the combined fragment may be merged in any sequence once the event sequence in each operand is preserved.
<code>seq</code>	Weak Sequencing results in the ordering of each operand being maintained but event occurrence from different operands on different lifelines may occur in any order. The order of event occurrences on common operands is the same as the order of the operands.
<code>strict</code>	Strict Sequencing imposes a strict sequence on execution of the operands but does not apply to nested fragments.
<code>neg</code>	Negative describes an operand that is invalid.
<code>critical</code>	Critical Region imposes a constraint on the operand that none of its event occurrences on the lifelines in the region can be interleaved.
<code>ignore</code>	Ignore indicates the message types, specified as parameters, that should be ignored in the interaction.
<code>consider</code>	Consider states which messages should be considered in the interaction. This is equivalent to stating that all others should be ignored.
<code>assert</code>	Assertion states that the sequence of messaging in the operand is the only valid continuation.

Communication Diagrams

- Hold the same information as sequence diagrams.
- Show links between objects that participate in the collaboration.
- No time dimension
 - sequence is captured with sequence numbers.
- Sequence numbers are written in a nested style (for example, 3.1 and 3.1.1)
 - indicate the nesting of control within the interaction that is being modelled.

Communication Diagrams



Model Consistency

- The allocation of operations to objects must be consistent with the class diagram and the message signature must match that of the operation.
 - Can be enforced through UML tools.
- Every sending object must have the object reference for the destination object.
 - Either an association exists between the classes or another object passes the reference to the sender.
 - Message pathways should be carefully analysed.